

Beyond2D

DSPRO 2



Fabian Dubach, Timon Schmid, Yannick Frischknecht



Problem

Looking at apartments online:

- 2D images lack in depth detail
- Can distort space (wide-angle lenses make rooms look bigger)
- Don't always show how rooms connect
- Can't see all the different angles
- Only depth information on the floor plan



Solution

3D reconstruction of apartment:

- Virtual property tour
- You can rotate, zoom in/out, and explore from any angle
- 3D models let you grasp the dimensions
- Gives you a more honest representation

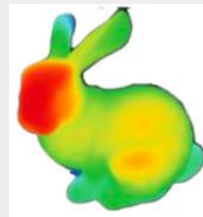


3D Representation Methods



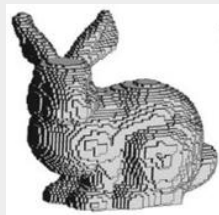
Point Clouds

Collection of 3D points in space.



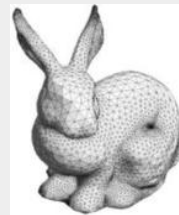
Depth Maps

2D image with per-pixel depth information.



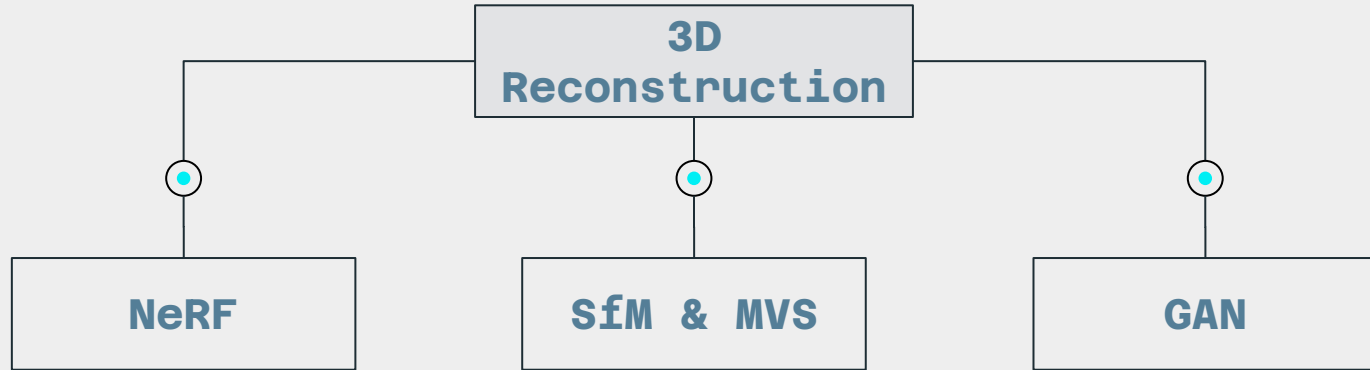
Voxels

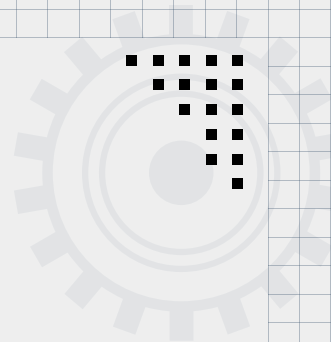
3D equivalent of pixels; space divided into small cubes.



Meshes

Represent surfaces using vertices, edges, and faces (typically triangles).





NeRF

Neural Radiance Fields represent scenes as **continuous volumetric functions** using neural networks.

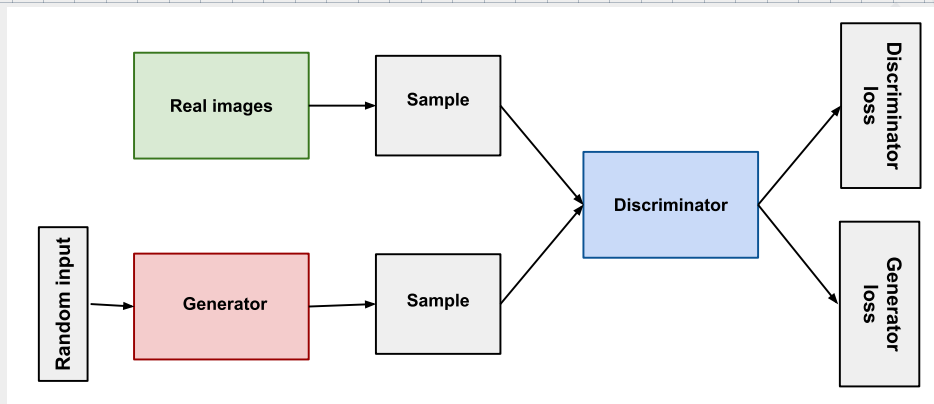
Given viewing **position** and **direction**, NeRF **predicts color and density**, allowing novel view synthesis through volume rendering.



- ⊕ High Visual Quality
- ⊕ Few Input Images
- ⊖ Computational Intensity
- ⊖ Slow Rendering
- ⊖ **Scene-Specific Training**



GAN



Generative Adversarial Networks use competing **generator** and **discriminator** networks to create **realistic 3D reconstructions** from images.

Given input **image(s)**, GANs produce **3D geometry** through adversarial training, allowing novel view synthesis and model creation.

⊕ High Fidelity Output

⊕ Works with Limited Training Data

⊕ End to End Training

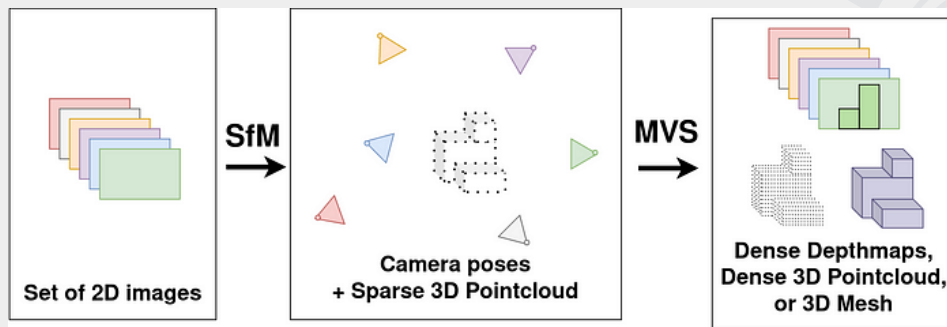
⊖ Unstable Training Process

⊖ **Resolution Limitations**

⊖ **Computationally Expensive**



SfM & MVS



Structure from Motion recovers **camera poses** and **sparse 3D points** from **multiple images**.

Multi-View Stereo then uses these camera poses to create **dense 3D reconstructions** by matching pixels across views.

- | | |
|-----------------------------|------------------------------|
| ⊕ Scene Scalability | ⊖ Quality Limitations |
| ⊕ Flexible Image Conditions | ⊖ Multi-Stage Pipeline |
| ⊕ Geometric Output | ⊖ Incomplete Reconstructions |
| ⊕ Hardware Efficiency | |

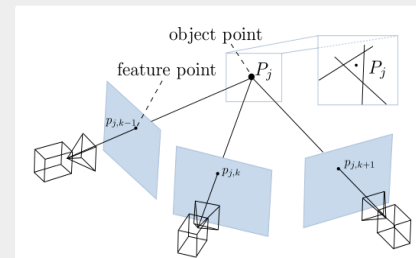
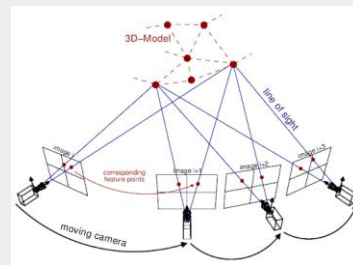
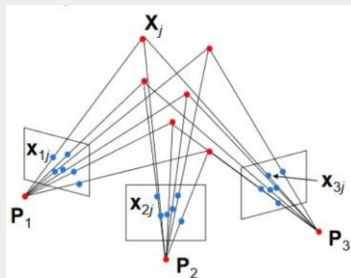
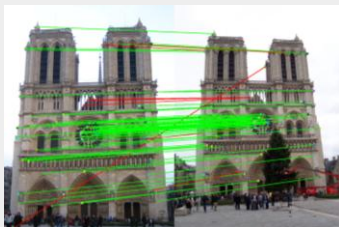
Structure from Motion (SfM)

Feature Detection
& Matching

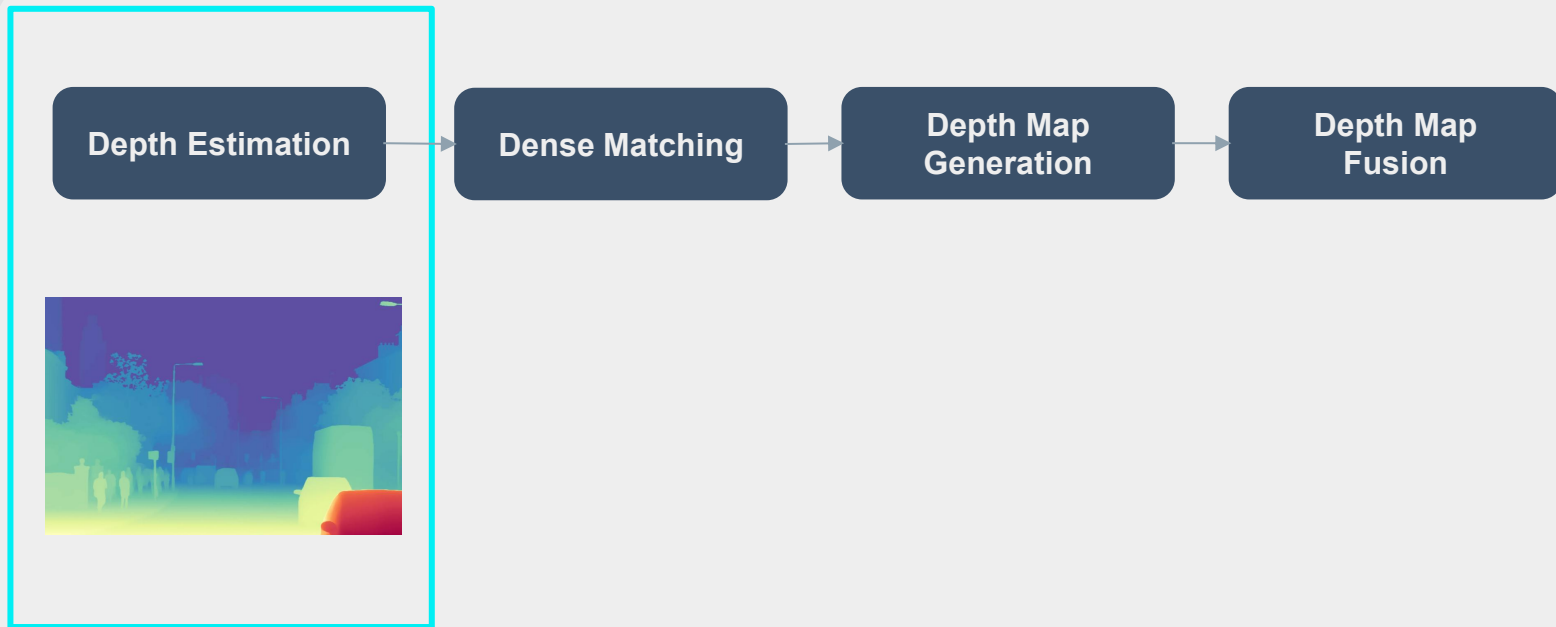
Camera Pose
Estimation

3D Point
Triangulation

Bundle
Adjustment



Multi-View Stereo (MVS)



Monocular Depth Estimation (MDE)

Predicts **depth of the objects** in a **single image**.
Provides initial 3D points for MVS without relying on feature matching.

There are two depth estimation subtasks:

- **Absolute** (measured depth)
- **Relative** (depth order)

Loss Function:

- Penalizes only difference in structure, not scale or shift

$$L = \frac{1}{HW} \sum_{i=1}^{HW} | \hat{d}_i^* - \hat{d}_i |$$



MDE Models

- **MiDaS**: Hybrid CNN-Transformer Architecture
- **DepthAnything**: Pure Vision Transformer built on DINOv2

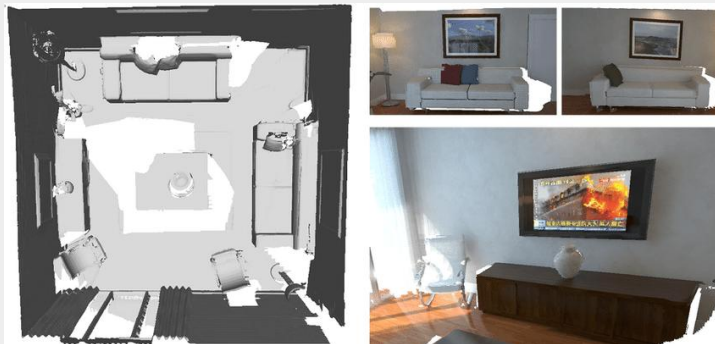
Model	AbsRel ↓	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
Midas	0.1464	0.8446	0.9601	0.9816
DepthAnythingV2	0.1755	0.8862	0.9719	0.9864

AbsRel (Absolute Relative Error): lower values indicate better performance
 δ_1 (Delta 1 Accuracy): Higher values indicate better performance

Data – Living Room

The ICL-NUIM Living Room dataset offers a single photorealistic 3D scene of a living room, captured via four distinct RGB-D camera trajectories.

Trajectories:	4
Total images:	4'601
Image size:	640 x 480



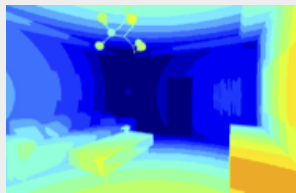
Data - DeepFurniture

A large-scale dataset for furniture understanding, featuring photo-realistic **rendered indoor scenes** with high-quality 3D furniture models.

Total scences:	24'000 photo-realistic rendered images
Total furniture instances:	170'000 annotated instances in scenes
Unique furniture identities:	20'000 3D models with preview renderings
Furniture categories:	11
Furniture styles:	11



scene



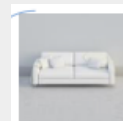
depth



segmentation

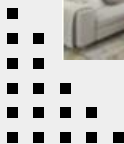


bounding box

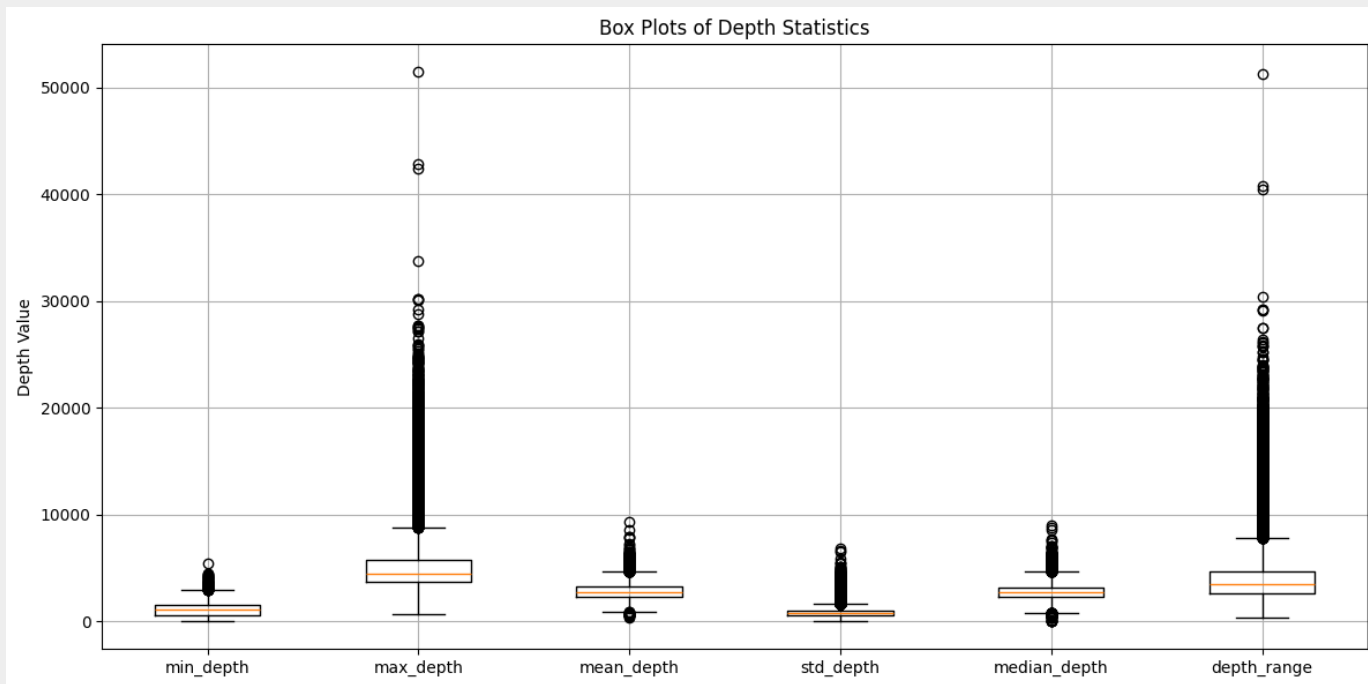


Id: 001
Category: Sofa
Style: Modern

identities

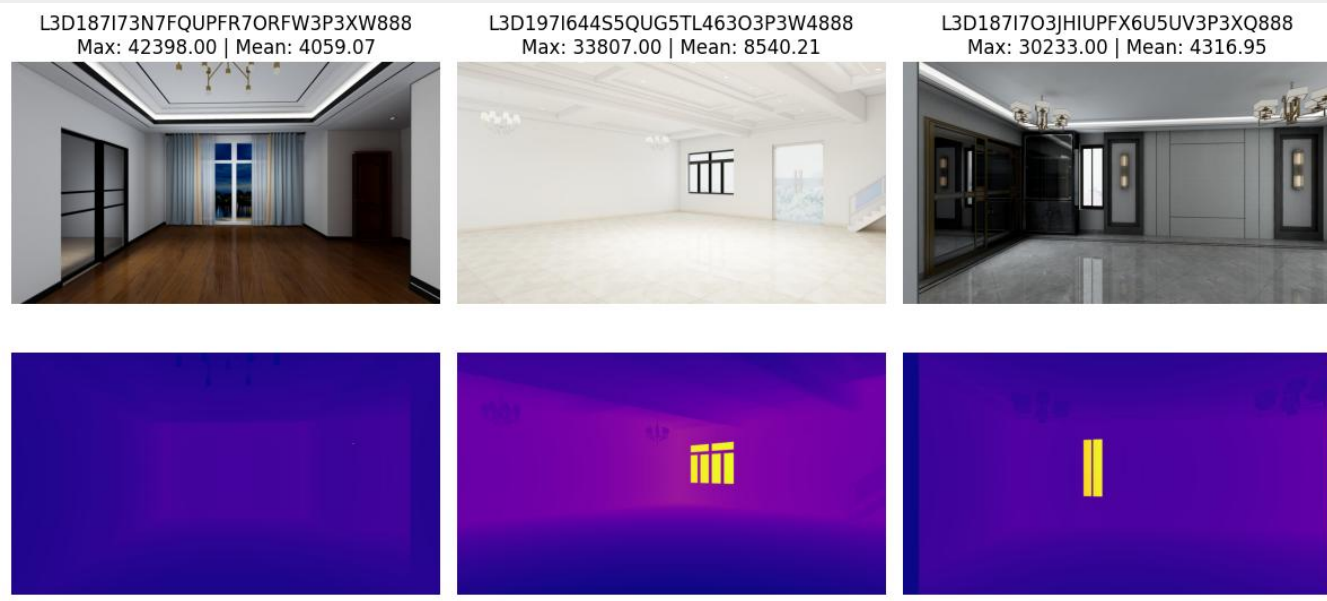


Data Analysis



Data Analysis

- **Windows** often produce extreme and inconsistent depth values
- Few **isolated pixels** show unrealistic depth values



Data Cleaning

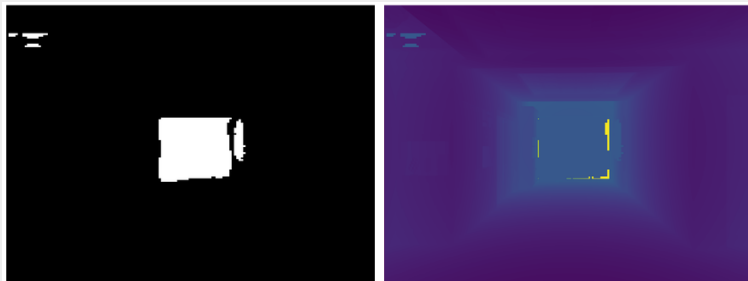
Threshold-Based Filtering

Removing depth maps with values exceeding 20,000 units (20 meters)

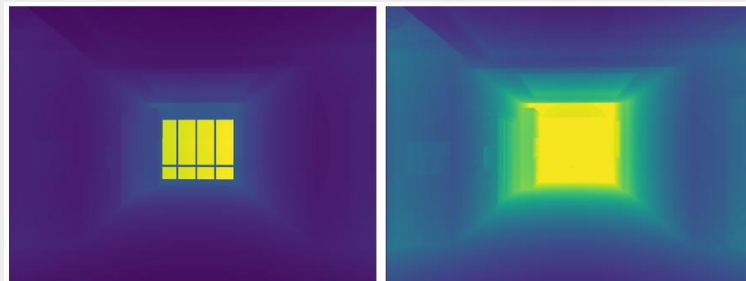
In-Image Depth Cleaning

Instead of removing data, we improved overall quality by directly correcting depth maps.

Semantic Segmentation



Direct Depth-based Thresholding

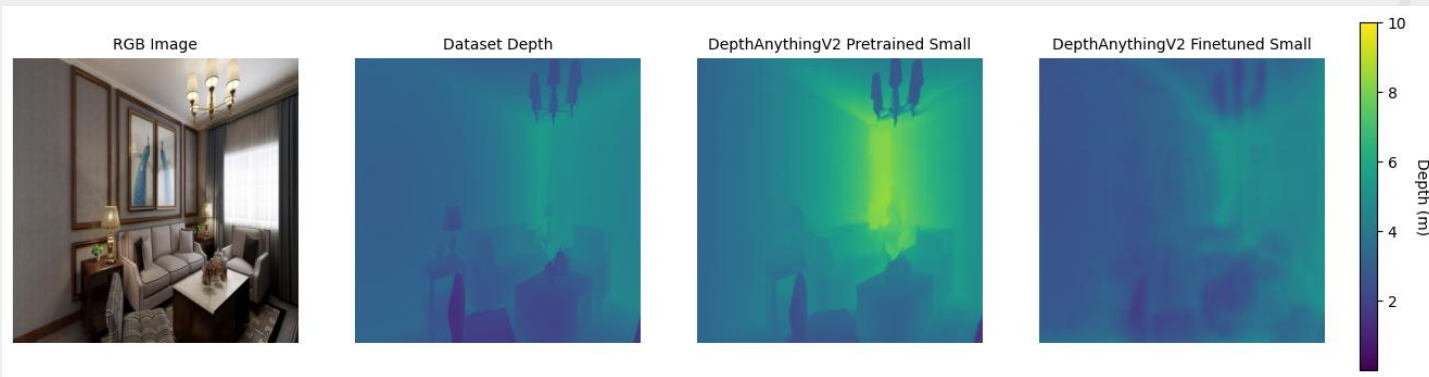


Results

Model	Encoder	AbsRel ↓	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
Metric DepthAnythingV2	ViT-S	0.9090	0.1524	0.5477	0.8142
Beyond2D - Metric DepthAnythingV2	ViT-S	0.2230	0.8375	0.9561	0.9811
Metric DepthAnythingV2	ViT-L	0.8728	0.1738	0.5994	0.8407
Beyond2D - Metric DepthAnythingV2	ViT-L	0.1309	0.9080	0.9687	0.9848

AbsRel ↓ (lower is better), and the threshold accuracies δ_1 , δ_2 , and $\delta_3 \uparrow$ (higher is better).

ViT-S Encoder



ViT-L Encoder



Interpretation

Corrects Scale Overestimation

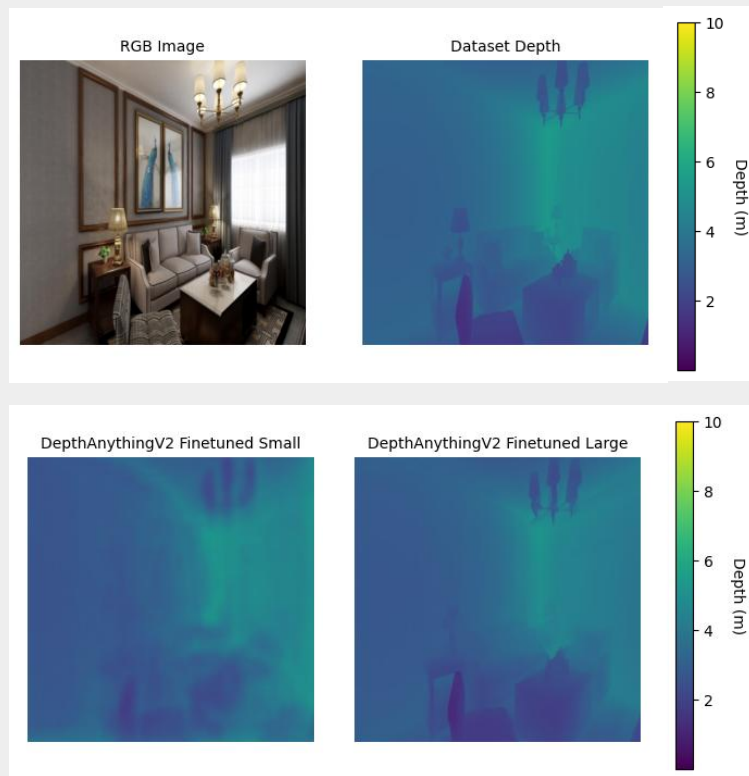
Pretrained models capture structure well but tend to overestimate absolute depth.

ViT-L Outperforms ViT-S

The ViT-L backbone benefits more from fine-tuning, likely due to its higher capacity.

Loss of Fine Detail in Deep Regions

Some deeper or complex regions still lack in detail.



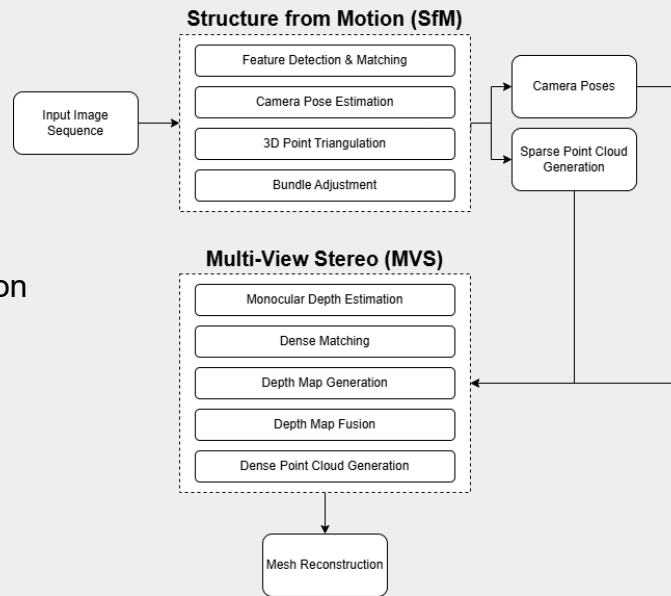
Model Integration

Goal:

- Improve 3D reconstruction quality of an existing pipeline

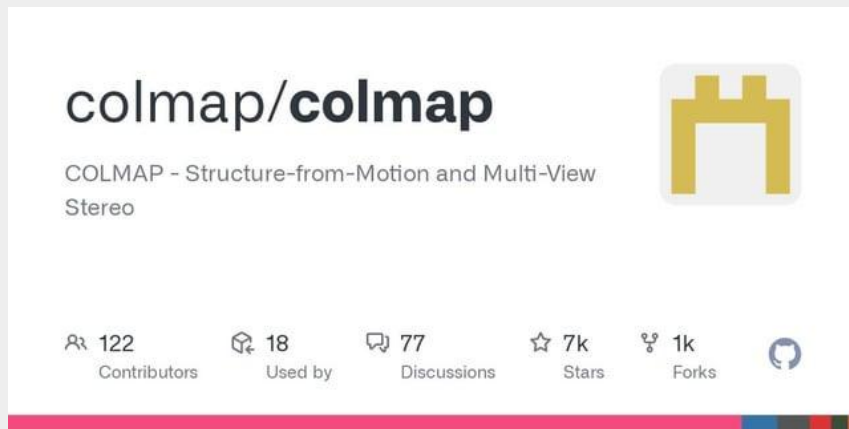
What we were looking for a SfM/MVS framework that...

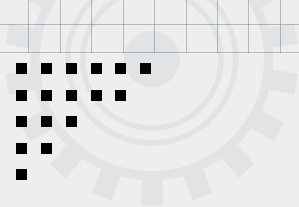
- ...has a suitable 3D reconstruction pipeline for model integration
- ...allows component swapping
- ...doesn't require camera information



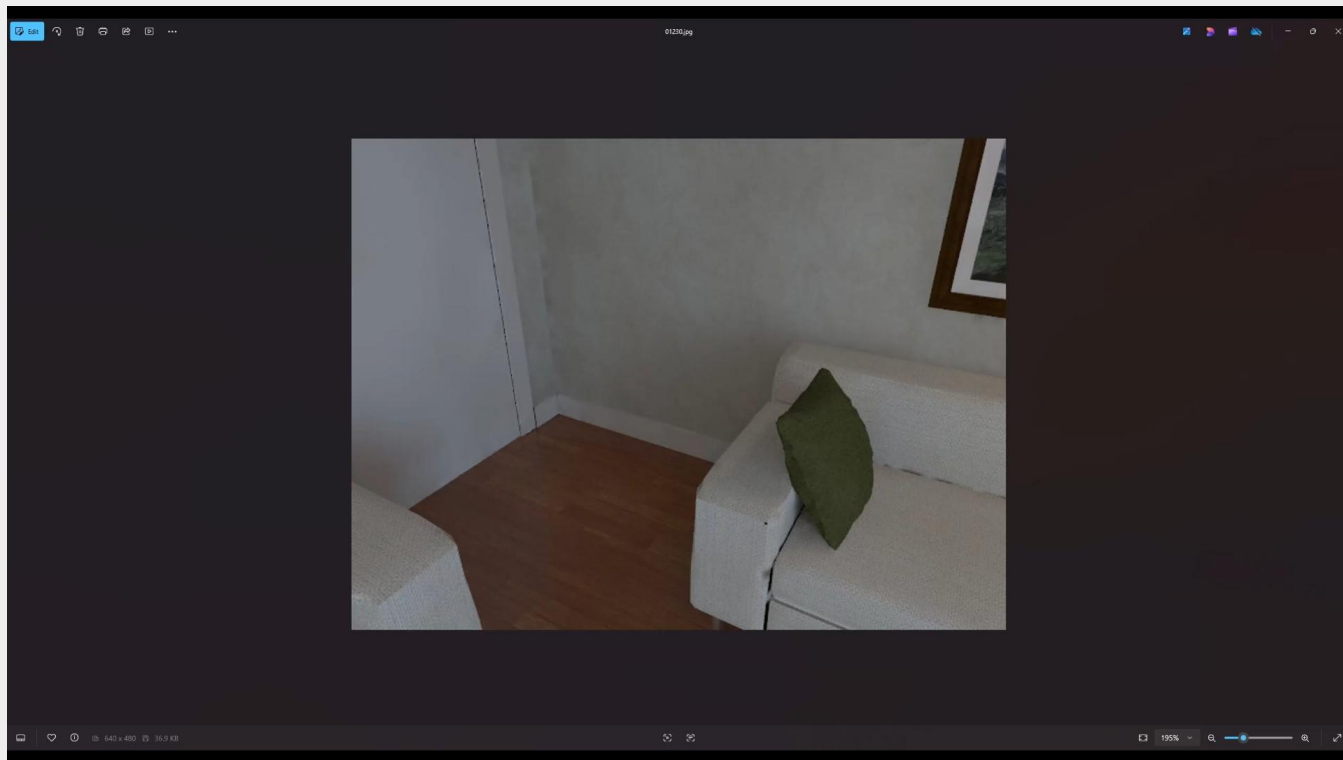
COLMAP

- 3D Reconstruction Toolkit (SfM/MVS)
 - Industry-standard tool for SfM/MVS
 - Offers wide range of features
 - Modular design (component swapping)
-
- **Input:** Images
 - **Output:** Camera poses, sparse point cloud, dense point cloud



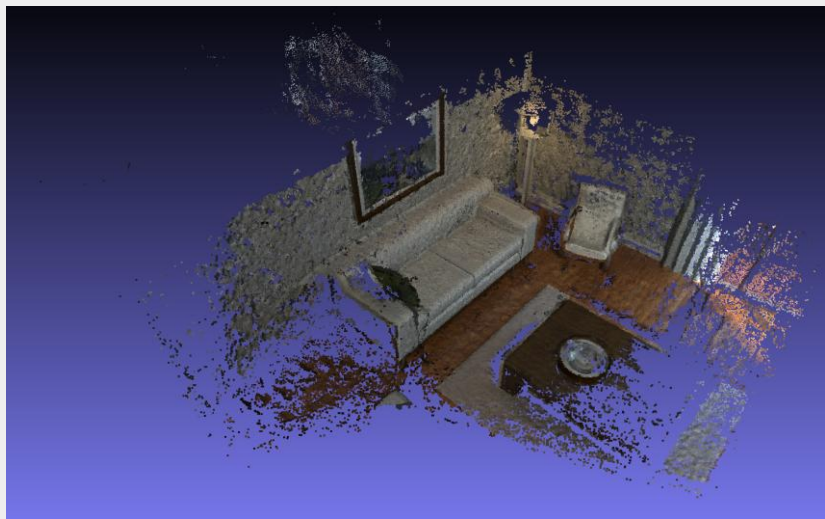


Demo



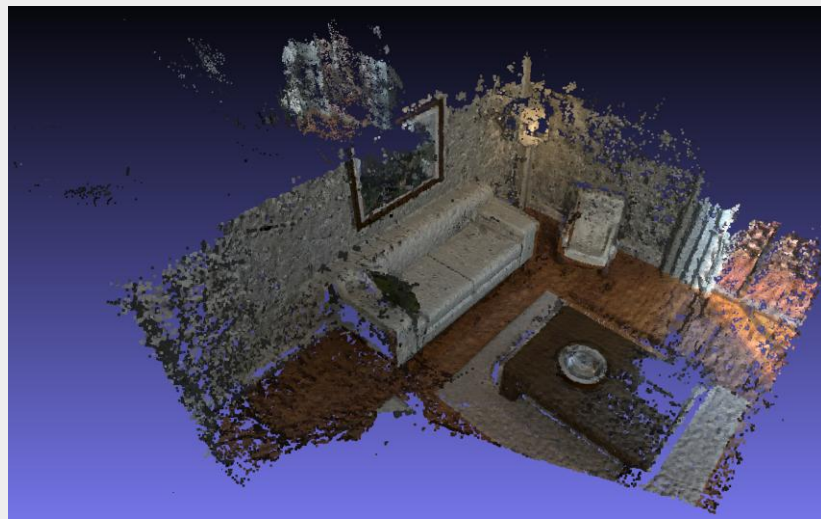
Comparison

Traditional COLMAP



255'566 points

DepthAnythingV2



524'738 points

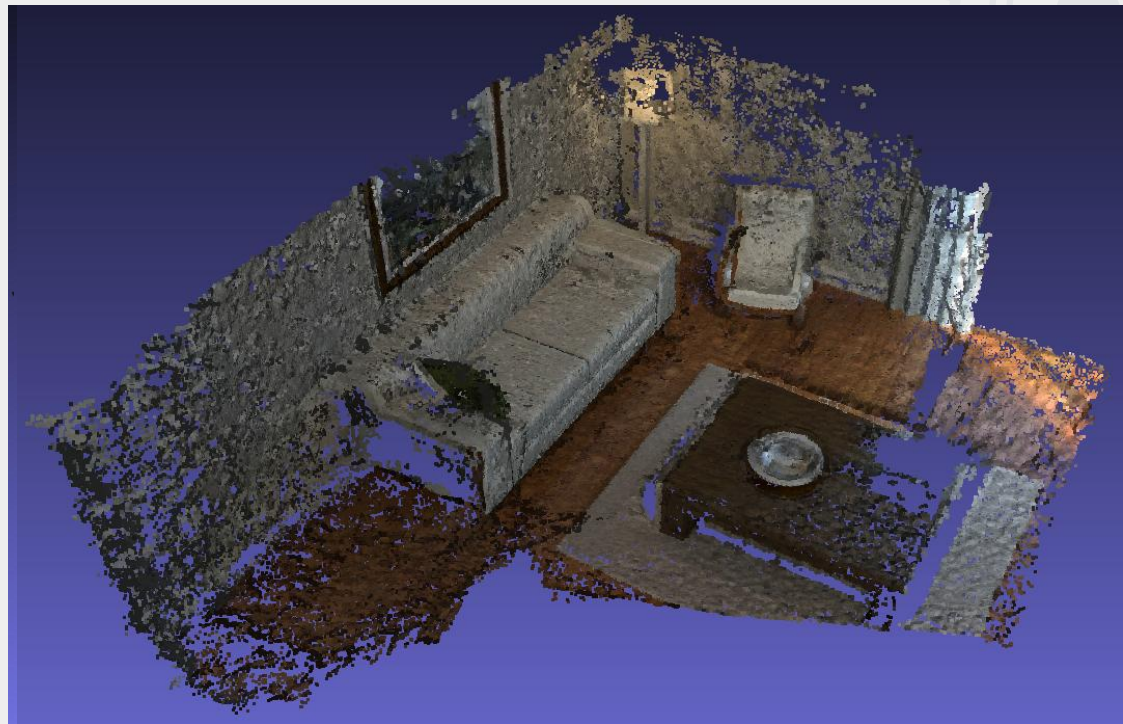
Model Cleaning

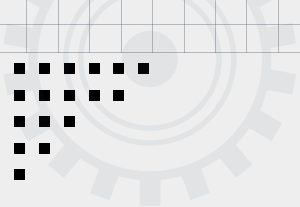
Python-based cleaning:

- Remove statistical outliers
- Remove radius outliers

Manual cleaning:

- Remove points from hand





Creating Mesh



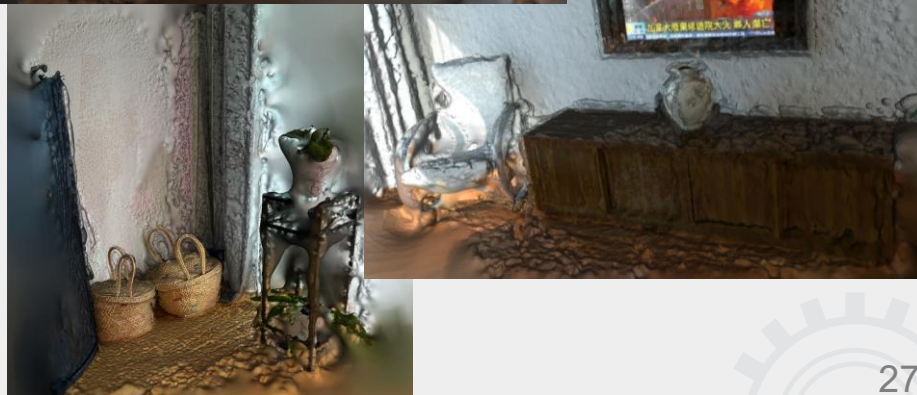
Multi-Scene Approach

Approach:

- Reconstruct each room separately, then integrate
- Align rooms using architectural features (walls, doorways, floor plans)

Benefits:

- Avoids single-pass limitations (occlusions, alignment errors)
- Overcomes processing constraints (too many images)
- Maintains high geometric fidelity at room level





COLMAP Problems



Large Image Sets

Becomes unstable with extensive image collections and inconsistent feature matching.

Image Spacing

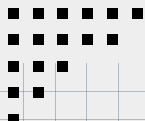
Requires close viewpoints with sufficient overlap for reliable feature detection.

Sequential Matcher

Single poorly matched image can halt the entire reconstruction process.

Parameter Tuning

Many tunable parameters make optimal pipeline configuration difficult.





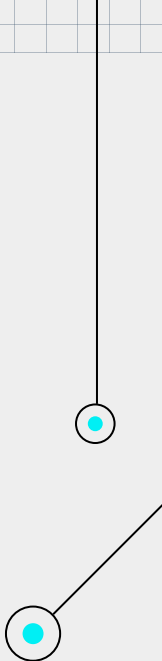
Conclusion



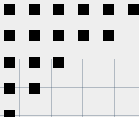
Successful Pipeline Integration

- Integrated our fine-tuned model into Colmap
- Denser and more complete point cloud

Significant Performance Gains

- Improved Accuracy
 - Decreased Absolute Relative Error
 - Corrected Depth overestimation
- 

Trade-off

- Loss of fine spatial detail
- 

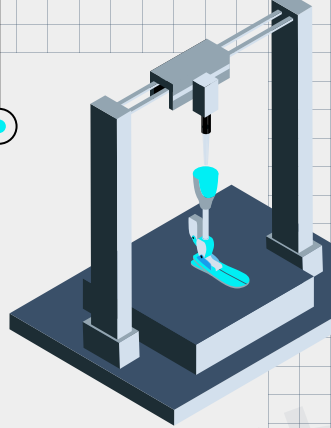
Future

COLMAP Finetuning

- Further boost point cloud density
- Reduce noise

AI Model in SfM

- More robust feature tracks
- More accurate camera poses
- Higher quality sparse point cloud



The background is a dark blue field with white geometric patterns. On the left, a series of parallel white diagonal lines form a partial rectangle. Scattered across the field are several white circular nodes, each with a smaller white dot in the center. These nodes are connected by thin white lines that form a network or circuit-like structure. Some lines are horizontal or vertical, while others are diagonal, connecting the nodes in a sparse, abstract manner.

Questions?